

Executive Context: Knowvia was developed to eliminate operational friction caused by informal communication tools (WhatsApp, Telegram) in tech hub internship programs. It delivers an isolated, role-governed progressive web workspace for **flexible class scheduling, screened learning materials, assignment evaluation, real-time group chat, and deep-linked announcements** across distinct departments.

5 / 5

ISOLATED WORKSPACES

42 / 42

AUTOMATED TESTS PASSED

100% Free

ZERO-COST CLOUD TIER

100% Clean

DEMO DATA PURGED

FUNCTIONAL MODULES DELIVERED

FULL IMPLEMENTATION

- Admin-Controlled Email Onboarding:** Public registration replaced with SHA-256 encrypted, single-use email invitation tokens (SMTP), user directory, and safe soft-deletion.
- Assignment Management & Grading:** Replaced bloated project boards with an academic workflow (Submissions, In-Review, Approved, Revision, and Tutor Feedback).
- Class Scheduling & 24h Cron Reminders:** Interactive scheduler with automated background node-cron reminders dispatched 24 hours prior to sessions.
- Real-Time Chat & Deep-Linked Announcements:** Department-scoped Socket.io chat rooms, global admin broadcasts, and auto-generated announcements that link directly to content.

SECURITY, ARCHITECTURE & QA

PRODUCTION READY

- Enterprise File Screening:** Multi-tier validation checking binary magic bytes against declared MIME types, blocking executable extensions, and enforcing 25MB limits.
- System-Wide Initials Avatar Engine:** Deprecated profile photos to prevent storage bloat and protect privacy, using a canonical deterministic initials engine.
- PWA & Background Web Push:** Service worker caching, installable manifest, and VAPID Web Push notifications delivered even when the browser/PWA is closed.
- Database & Zero Demo Footprint:** Migrated to Supabase PostgreSQL, permanently purged all mock accounts (David Kim, Marcus Chen), and isolated test environments.

TECHNICAL ARCHITECTURE & DELIVERABLE STATUS

VERIFIED SEPTEMBER 2026

Layer / Component	Technology	Key Functionality	Status
Client Interface	React 19 + TypeScript (Vite)	Clean light theme, vanilla CSS tokens, role-based dashboards, responsive PWA	COMPLETE
Backend API	Node.js / Express.js (TS)	RESTful controllers, Zod validation, JWT session handling, SMTP service	COMPLETE
Real-Time Layer	Socket.io (WebSockets)	Live department channels, active typing indicators, real-time alert broadcasts	COMPLETE
Data & Persistence	Prisma ORM / Supabase PG	5 isolated departments, relational models, soft deletion, migration ready	COMPLETE
Security / Storage	Magic-Byte Validator / Multer	Executable blocking (.exe, .sh), sanitized file headers, UUID prefixing	VERIFIED
Push & Reminders	Web-Push (VAPID) / Node-Cron	24-hour automated class reminders, personal notification bell drawer	OPERATIONAL

Current Build Health: 0 TypeScript compile errors; 42/42 Jest test cases passed; Clean working tree on main.

Next Milestone: Final production deployment to Render (API) and Vercel (PWA) with live Supabase storage bucket linking.